ActInf GuestStream #038.1 ~ Max Berg "Oversampled and undersolved"

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hello and welcome it's March 9th 2023 we're an active guest stream number 38.1 here with Max Berg thank you Max for joining this is going to be great really looking forward to your presentation and discussion so appreciation again for joining and looking forward to what you'll share with us today yeah thank you Daniel for having me um as I was introduced my name is Max back I come from the University of Marburg and I'm a clinical psychologist I want to do a little bit of impression management first so Marburg is next to Frankfurt in case you don't know it's a small or medium city in in Germany and regarding me I'm interested in the intersection of Clinical Psychology neuroscience and philosophy that doesn't mean that I'm an expert in all of these fields of course I'm I have a lot of knowledge in Clinical Psychology I would say and the rest is more of my hobbies and so it's important that you keep in mind to expect a clinical perspective today I'm not an expert on like uh Bayesian mathematics or anything like that I know a little bit about it and I've read uh books about it and try to keep myself updated but it's not my expertise my expertise is Clinical Psychology and one I want to present today is like a paper about depressive rumination but we will see that it's not important that it's depressive rumination and rumination is a much broader construct here and um we want to [Music] re-rephrase like the Press of rumination from an active inference perspective and this is one of what we want to talk about today yeah so um I want to start with a little quote and it's rumination is the coin of my realm interiority breeds interiority and we will see that this might be literary true and this is why I think it's a good quote to start with so what is rumination it is uh defined by the APA as obsessional thinking involving excessive repetitive thoughts or themes that interfere with other forms of mental activity it is a common feature of OCD and generalized anxiety disorder I actually don't know why these two disorders were chosen first but well here we go I just quoted them here I would add a major depression to the list and at least in my mind measure depression is the Prototype disorder for rumination but sure OCD and generalized anxiety disorder also has a lot of rumination going on and here you can see a first really important Point namely that rumination is its runs diagnostic marker right it's important for multiple uh psychiatric conditions and yeah as said it's it's an important Trends diagnostic risk factor and it can predict actually depression scores and subsequent measurement time points and this is what makes it interesting there was like a Grant and small Grant proposal by I believe the welcome trust in in the UK where they gave grants for writing systematic reviews about Trends diagnostic topics in mental health and unsurprisingly rumination is one of the topics where they gave or where they wanted to have systematic reviews for which yeah they can see that it's an important topic where people really believe that it has some um has some utility clinical utility and unfortunately it is also the case that it's associated with worse treatment response for Psychotherapy and also for medication treatment yeah and let's make it more complicated shall we um the rumination is sometimes used interchangeably with the concept of repetitive negative thinking and sometimes repetitive negative thinking means rumination and worrying um but we will use the definition of rumination in its broader sense today so when I talk about rumination I also mean worrying some of the time I just wanted to make this clear at the beginning so I use it in the older broader sense so to say yes so why would we need a new uh concept for rumination well reinforcement learning I would say reinf struggles to explain the constant continuation of ruminative thinking to make it short why should you continue with doing something that is not giving you like pleasure or any good feelings in the short or the long run and yeah why is something continued despite not promoting um action and also leading to depressed mood this is a question at hand it's not an easy question to answer I would say and this is why we would say that we have to not only take exploitation learning into account but also take exploration learning or surprise reducing learning or free energy minimization into account however you want to call it the important part is um you have to take into account that not all learning is done to uh in the sense of like reinforcement learning but there's also learning that we do to know more about the world and if this goes wrong rumination can occur and this is how we want to rephrase it today yeah so this is a roadmap for today we want to reconceptualize rumination as an extreme case of failed mental problem solving we want to present a model of rumination using active inference terminology so this will be a conceptual model we don't have the mathematics to back it up at the moment I will talk about this limitation at the end again and we will also discuss neurobiological correlates shortly to then continue to point towards treatment implications of the model so the next part is like I want to talk a little bit about problem solving and mental problem solving and I've had a classic quote by Richard

Dawkins that I really find interesting in this context it says survival machines that can simulate the future are one step ahead of survival machines who can only learn on the basis of over trial and error the trouble with over trial is that it takes time and energy the trouble with overt error is that it's often fatal stimulation is both safer and faster yeah and I think this is really interesting and we can talk about also like this evolutionary perspective later I have really broken down into simple terms first okay so when I want to solve problems uh using my mind what do I need to do well at first I need to think about what to do next what do I want to think about so to say the um and we have called this step candidate sampling so it's meter policy basically so what do I even want to think about before thinking about what are important tasks of today for example and then the next step is policy evaluation or if I think about that what are the expected results if I tell them about this concept will they understand my talk today for example and then let's say yes I hope this that this is the case then we have to do another step um and do like the actual problem solving okay now I'm convinced that this might be the right way but how to do it how to actually solve the problems which are the steps necessary and these three steps we will consider over and over again today and we have in the paper a neat little example that happens to me a lot that's um one of the reasons I really chose today imagine you lost your key and now um what we typically do is at first we check the usual places or have her put it there no have a maybe put it there no and after a while if this doesn't work my policy changes typically and I think okay um where did I see my key the last time that I saw it and trace it back and maybe this works and I find my key back and a few days after I have the same problem again because I already lost my key again and what I may do after a few instances of this behavior is like okay I need to change something fundamentally in order to get my key back faster and so in Bayesian learning terms at the beginning uh my expected prediction error minimization is just search the usual places it might work okay after a while uh the mental policy okay why do I keep losing the key and how do I solve this problem might lead to a higher expected free energy minimization or prediction error reducing properties and this is why I switched the policy right after a while so this meter policy step decides how to go or how to go about a given problem in this case how to find my key good this is really important we cannot search the interactively large probability distribution for this step it's not possible to remember all instances where I lost my key and to compare like promising strategies based on the whole distribution I'm not able to do this and so uh it's no machine no living being we have to find fitting samples so to say and fitting means um it's usually based on heuristics what did I do with the last time when the key was gone how can I find it this will become important later because these heuristics can run into problems they help us um like make computationally intractable problems uh solvable but these heuristics have also their downsides as we're going to discuss then step two now uh policies are then selected if they provide a trade-off between pragmatic and epistemic value and pragmatic is okay they I have a good chance of finding my key back to stay on the example in epistemic is I learn also something about finding keys or understand the world better and uh this is also important whenever I choose a specific policy um High opportunity costs may arise what I mean with that is like when I pursue like a searching process for example for finding my key I cannot continue preparing my talk or something like that so every decision I make and um has necessary costs because I cannot pursue other goals in the same time and this is also becoming uh going to become important so in order to be chosen by an organism each a chosen policy has to outweigh the resulting opportunity costs uh otherwise it's not adaptive right okay and now the last step um the mental problem solving good I now decided for one strategy of finding my key and now I have to actually pull it off so where to search what to do next which sequences of movements do I have to perform uh and how to go about it best and um typically in different strands of literature not just in active inference I need to perform something that is called it research you know this from chess playing typically or for class computers like they do it a lot better than we do but they look okay if I do this move um and he does this move then I can do this move and in all move that follow it I have a good position and it's again not possible to search the whole tree because the more branches it becomes um the more branches are data branches of these branches again so uh the tree growth exponentially so I need to have a strategy of not searching through all the branches it's called the pruning strategy in the case of Bayesian learning it's often Occam's window meaning that branches are not searched through if they don't promise a prediction error minimization that is sufficient to pursue searching them this can also lead into problems because it's a branch at first doesn't provide enough um prediction error and minimization it will not be searched through despite the fact that it could have some prediction error minimization further down the road so to say but we have to prune again because it's not possible to search the whole distribution so with these three steps in mind we can have the following model so we have the candidate sampling uh how to go about the lost key so to say policy evaluation if I do the strategy have I do I have a good chance of finding the key and learning something about the world and mental problem how to actually pull the strategy off and I want to say one really important thing and this is an increase in mental problem solving is a normal process in a lot of situations right so for example if I lose a person that is really important for me it's normal that my general models had decreases like a lot of my strategies for getting social supports don't work anymore because the person's gone and so uh in in in Bayesian terminologies prediction errors increase after such an event and this is a normal process um and if like my general model if it declines this was linked in the active inference literature to bad mood or to yeah feelings of sadness for example and in such cases typically you think more about a fundamental things in your life everybody knows it and has experienced it if you don't feel well you tend to think more and to try to get into a better situation again and hopefully what happens then is okay I uh for example I lost someone my social support system is breaking together um or it's breaking apart it's falling apart so after a while I searched for people that I like and that can form a new bond with um and hopefully after a while my prediction errors become smaller again but therefore I need to take action I need to think about what to do with my life um how to get to know new people in this example to like reduce my um prediction errors in this case regarding social support and then I have to pull it up I I have to go to certain places to meet people and hopefully after a while of sadness and difficult times um my general model fit increases again but this is not always the case as we know uh want to discuss next so I have yet another quote you will get to you get to use to quotes today sorry uh the next quote is about rumination and it says uh shut up she tells her monkey mind please shut up you pick off nips press of bruises counter of losses Sierra or failures collector of grievances future and past and now we want to discuss what happens when this doesn't work so it can happen that despite trying agents and organisms don't find good approximations for important problems in their life right and if this is the case the prediction errors regarding these problems stay high because the surprisal doesn't get resolved and um despite trying the organism tries over and over again prediction errors say hi surprisal doesn't get uh resolved and the prediction error doesn't reduce again and the problem stays and what we now want to do is discuss reasons for why this happens in depressive and at first we have to say Okay a typical rumination problems have a high complexity with uncertainties at many hierarchical levels if you think about problems for example like why is the world such a cruel place for example this could be a nice rumination problem you would have to gather a lot of evidence from different strains it's really hard to actually disprove such a claim or to prove such a claim or even to say what it means and this is why there might be no efficient solution for such a question it's not like navigating a maze or something like that but it's much more complicated and in addition sometimes like we human beings have a very high Precision demands for solutions for example after I may have survived a traumatic event I might be really reluctant to go to similar looking places again and I have to be really really sure that the new place uh where I don't want to go is really different from the place where something bad happened to me and if I'm not sure and sure means really through a high Precision I won't go there these complexity demands can also make an efficient approximation to a problem improbable this might be one reason why problem solving isn't functioning in the case of depressive rumination another reason might be uh over pruning and missing action policies so as I said beforehand like pruning in order to uh function properly or let's say this way pruning is a good strategy when the parts of the search tree that I removed so to say don't yield like at the beginning small losses and at the end large gains because if it's like that the three branch might be pruned away despite actually helping the organism to reduce surprisal but the organism will never find out because it gets put away the next thing is also action policies can all only be thought through if an agent knows about them we know about a lot of depression patients with a lot of childhood adversities that actually don't know a lot of strategies for navigating social environments um it's not their fault it's the consequence of childhood adversities but nevertheless this is a problem if I don't know the solution for a social problem I cannot approximate a solution because action policies are missing the next problem might be that after many failed attempts of problem solving I might not even have a reason to engage in deeper instrumental learning what I mean with that I go back one slide um after a while I have an idea so okay why is my life becoming so shitty uh it is because I don't engage with other people anymore I just lie in my bed and feel sad okay and now normally there would be a step where I think about how to engage with other people again but if I have for example missing action policies really bad mood and not a lot of strategies I might think this through shortly and think well I can try but they won't like me anyway so instead of deep mental problem solving I uh jump back to step one which is why uh do I have so many problems with other people and there you can see your ruminated circle is forming in no deeper engagement is possible anymore and if this happens my prediction errors stay large and more candidate sampling occurs okay I cannot solve this by meeting other people but maybe I can do it on my own no I feel so sad and lonely you see where this is going I do not engage anymore in deeper problem solving but instead thinking through a lot of policy candidates and none of them is like giving me confidence that I can solve my problems so I um stay at the same meter policy selection step so to say and in consequence I have less instrumental thinking and less instrumental learning and this also has consequences um for example the pragmatic value of policies are estimated to be low due to low model fit and this even becomes worse over time so if I think I cannot do it I will not train it if I do not train it I become even worse at doing it and so I want I will it becomes more and more improbable that I try because my action policies become more and more undertrained yeah this is another really important and the next problem is the opportunity costs because illuminate a lot so now we are at a meter level I don't do much else and because I don't do much else my models it further declines and yeah the rest we talked about on the last slide um ruminative States may be learned as agents default response for quote-unquote dealing with problems after a while and now we have a vicious cycle here the next uh thing that I want to discuss is um how neurobiology might play a role yeah and I have had a little joke of Phil Colette ripped off for my slides here um and uh quotation by Carl Friesen that says an agent does not have a model of its word it is a model in other words the form structure and states of our embodied brains do not contain a model of the sensoriums they are that model and on the right you can see like uh Yale School of Management guy that actually really embodies this Principle as he's clearly showing on his backpack okay Jokes Aside what is first to be said uh the default mode network is going to be important for our discussion here the default mode network is a large scale network with media frontal temporal and parietal region from the brain it's important for introspection self-referential thinking and autobiographical recoil so in short it's important for everything mental problem solving and rumination related or a lot of these things and this is why it's unsurprisingly going to play a role um to be more precise the frontal courtesies seem to be the main driver of a memory recall in the default mode Network so if the default mode Network tries now I'm intramorphophizing it um to recall a memory so where do I find the key for example it uses the frontal regions to be the main driver of this process it is the case that meter studies found a stronger coupling and the hyper activation in default mode Network structures to be correlated with with ruminations and they're actually two meter analysis that found this and it is also suggested that a coupling with a subgranule cortex might contribute to the negatively valence Nature's rumination MDD so we know okay dmn is important for problem solving and for rumination and it's also found meta-analytically that it indeed plays a role for a rumination or at least it's correlated in addition meta studies also found evidence for a reduced frontal parietal control Network within network connectivity in major depression and this is also interesting because as we discussed our idea is basically that rumination means I get stuck in abstract thinking and don't use my instrumental learning capabilities as much and so it makes sense that in MDD we find reduced control Network or frontal parietal control network activity note by the way that this part of the evidence line is more indirect than the part before right I don't want to gloss over this so while we have here a direct correlation with rumination in the case of the default mode Network here it's more MDD in general this is why we also wrote it much more careful in the paper because it's a more indirect line and I don't want to gloss over that it is further more known that the connection between control networks and default mode networks is seem to be reduced and interestingly these two networks have to connect in order to solve complicated problems they actually study about that and these studies suggest that if the problem gets harder for an individual it uses more of the default mode Network to do the meter thinking and then the control networks to do the problem solving and the interplay between the two is really important and okay so in conclusion and what we think could be the case is that the default network is hyperactive and Hyper connected because it does the policy candidate over sampling that we talked about before and the reduced mental problem solving or um due to the policy oversampling is reflected by the reduced resting state activation in the hypo connectivity of the control networks and also of integration hubs like the insula then a short citation it will be the last one and it's the existentialist model and it says simply existence precedes essence and it wants us to say um in the first instance we are being that get thrown into the world but we have to do something with our lives in order to flourish and give it give ourselves reasons to live and to succeed and I want to talk about a little bit about like how can we get out of ruminative loops and what might be old idea and new ideas at first I want to say there's good evidence for the effectiveness of excrete of existing psychotherapeutical treatment regarding depression in general and it's also regarding rumination so we also have like meta-analysis uh specifically analyzing rumination however um we have substantial non-responder rates for Psychotherapy and also for current medication approaches we have a higher chance for relapse after depression is gone it will oftentimes reoccur in the life of patrons unfortunately and we also have substantial therapy dropouts so patients actually leaving therapy so it's really easy to say that uh psychological and pharmacological treatment needs Improvement so they are effective but they don't help everyone and there's a lot of room for improvement here and there are multiple ideas that we briefly discussed in the paper and the first is like a simple idea but could be quite effective and has also meter analytical evidence that it helps and this is like reduce memory interference and rumination by physical exercise in the everyday life of patients so doing more like especially endurance exercises actually helps reducing rumination even when compared to active treatment like CBT so maybe the present patients don't only need a therapy to be paid for but also to have a personal trainer that might be paid for and I only say this half jokingly this could improve the life of a lot of people and another idea is more medical it's psilocybin or ketamine was was shown in in studies to increase the sampling entropy of the brain and this could also be an interesting aspect because we've learned that the samples are getting repeated over and over again but um if we could have a idea for how to increase the variance of samples drawn when a probability that one sample is actually a good idea for thinking through increases however I have to say cautionary notes especially with um psychedelic drugs because uh it is a promising pharmacological Avenue probably but at the moment we have really small sample sizes it's hard to Blind for these drugs between groups I mean you know when you're tripping it's it's quite easy to know so this makes rcts difficult um and we know from a lot of questionable research practices from this line of research unfortunately I don't want to say that everybody in these fields are doing bad science it's not what I want to say but I know from a few cases where prerequisite protocols don't fit the results that were presented and stuff like this and if you combine all this it's actually quite hard to say whether these work really well so what we would need is really larger study uh not the small brain Imaging studies with 20 subjects more like 200 subjects would be a beginning um and we need the researchers to actually stick to predefined interpretations of their results so we can learn about more about this province promising Avenue okay and more ideas would be uh to change therapy a little bit and uh concrete idea is we could actually um add problem complexity estimations uh into therapy what I mean with this is like ask patients okay how much time you spent ruminating what typically are the results of these processes what are the costs of those processes so we could try to make the costs of the process more explicit to the patient and to give them a grasp on how much opportunity costs they invoke on themselves by the process it's not typically done at the moment but it could be an idea for improving CBT what is also important this is done in CBT what I now tell by the way but it's really important to keep it in mind we learned that we a good problem solving requires an intricate interplay between default mode Network and brains control networks so it's actually important to have optimal difficulties for Behavioral tasks given to patients if it's too difficult they will not do it and if it's too easy they will not learn anything from that nor epistemic value is mine so to say and both options are not good for reducing rumination and there's actually another experimental treatment that shows some success and it's called emotional flanker tasks in these tasks like patients learn how to shift their attention away from Bad emotions and this could also help like stop the ruminative process because we learn that depending on your mood you will have different heuristics in your brain about what to think about next and in the case of bad mutes you don't have the best heuristics so this is another idea for a treat for better treatment yes now I want to talk about a few limitations and then we can enter the discussion so one limitation is clearly that the model only discusses typical depression where Behavior breaks down however there are cases of agitated depression where behavioral output actually increases or to make it more abstract depression is a really heterogeneous subject and also rumination is really heterogeneous so we don't have a model actually at the moment that considers all individual trajectories so this is only a general scheme so to say but it will not fit every individual and the next thing is we describe plausible ways for rumination development but we do not know whether these Pathways play a causal role in MDD development I said briefly that uh when I have rumination at time Point T I will have more depression at time Point t plus one but we do not know whether this is the causal Factor but or only a coincidence in time you know what I mean so this is also an unsolved problem at the moment and the last problem which is also important for the active influence nerds in the community is it's not a mathematical model it's a conceptual model that we did and so the next step would be to make a mathematical model out of it for example like a partially observable Markov decision process or something like this um in order to be able to simulate it um and to check whether the behavior that we set it would come out actually come out so this is another major limitation of the paper yes I will skip over that I guess um we talked about this enough um so what I want to talk about shortly before entering the discussion is other projects that we want to do next uh with respect to actors in France and what we actually have done and it's now at under review at psychological medicine is uh applying the access inference framework to heterogeneous social experiences and behaviors and this is actually done in a mathematical precise way so there we have a hierarchical prompt uh or partially Observer the mark of decision process model and we use it for trying to explain like social withdrawal or other social problems in depression so it's a little bit rumination adjacent and it's more mathematicized a preprint is available but it's not yet through the revision process and what we also want to do is like simulate distorted social functioning in May Intel disorders um and also to compare these simulations or computer simulations based on active inference um to empirical data because this is also what we found in the literature to be missing we have the simulation studies right where we have like these simple games where active inference computer agents play uh like a trust game or something like that or where you simulate hallucinations and psychosis and these are really good studies don't get me wrong but what is missing in our view is a little bit combining these computer simulations with the behavior of human being to compare them with on within one and um one of the same design and this is what we aim to do but it's an ambitious project and we are not finished so this will take more work and another project that we actually will finish this year and we have done the literature screening is we have prepared like a systematic review of all active inference literature that is of interest in for psychological treatment and this will hopefully at least be a pre-print this year yeah so thank you these are my co-workers and different projects to drop a few names on the left of Matthias Feldman then Lucas kirschner and to be as cool with these are the guys where that I wrote the rumination paper with then there's Angelina cheese from a theoretical cognitive science groups and she does a lot of the computer models for our new projects uh so we need actually scientists that are better at programming than I am um to and she is doing a lot of the work here and then there's Philip Hartsock and bergett climb and with those two people uh I also collaborated on uh predictive processing model of PTSD which was also um published in neuroscience and biobehavioral reviews at 2020. yeah so with that I would like to thank you for listening um yeah we can discuss a few questions if you want now awesome thank you all right well thanks for the great presentation looking forward to the um discussion so just to sort of begin how did predictive processing and active become integrated with your and your colleagues research Direction were you looking for integrative Frameworks and came across this area or did this motivate you to study and enter into the clinical space it's a mixture of two things I would say one is like observations in patients um maybe start with that so my friend Tobias Uber got me interested in this stuff and he had a patient and this patient was like really uh had had to struggle with social phobia and what was interesting about him that he was discarding information that would challenge his world for you so to say right so whenever somebody said that he actually liked him this one said oh no he just pretends to say this because he wants to be a nice guy but in reality nobody likes me and this is what got us interesting interested in this line of thinking about information seeking for example how do organisms seek information to uh to run their lives but also how do organisms discard information uh and stay in a depressed or in an anxious state and this is I guess one of the reasons why we started to get interested in active inference and another reason um comes from my love or love from a lot of my colleagues um for simple explanations and I don't mean simple as mathematically simple because I don't get most of the mass to be honest fully but simple in the sense of we can have a only a few axioms and explain sentient Behavior with it we can computer simulate behavior in it and it actually produces interesting Behavior patterns that we know from the literature and this is the other reason why we got into it awesome so many directions and ways to go but one uh excerpt from the paper that that jumped out at me was the line between adaptive problem solving and self-perpetuating maladaptive ruminating how do we know that line and how does that play out when you're in the clinical context like what is the appropriate amount of thinking or overthinking how do we really find that line and come to know it that's a short answer that is we don't where the line is because I mean in principle maybe you find a good solution if you ruminate just two hours more right you never know but we have an answer in the clinical field in practice and it's like a pragmaticist answer and it only goes okay if you suffer a lot and don't get your everyday life sorted out then it becomes a problem and we have like measures for that so you can assess the global functioning level of a person and say can he continue to work healthy friends has he or she free time activities and actually fulfill her or him and this way you can at least say okay um has this person a good good level of functioning and if not does rumination impair this level of functioning and uh yeah this is a really pragmaticist answer sorry for that but this is how we do it in clinical practice totally totally makes sense um also in that kind of clinical area so I was curious other than rumination what other trans diagnostic risk factors what are some examples of those risk factors and how might they be linked together in a kind of holistic understanding or what what is the active and non-active way of thinking about the causes and consequences of these trans diagnostic risk factors yeah yeah one that jumps to mind is emotion regulation that comes up over and over again um so this is a risk factor also for depression it's a risk factor for borderline personality disorder but it's also a risk sector for violence so a lot of things are connected to to poor emotion regulation I have to think shortly to get an active inference answer out of that there's actually a paper a new one by my colleague Philip Herzog about like a predictive processing model of uh of borderline personality disorder where he describes in more detail so he is the expert there um but maybe we can just look it up before I make stuff up that's it not to can I just search it shortly and give this as an answer this is a internet extended cognitive yes because I don't feel confident enough to talk about the stuff that was not my main area of research so extended rumination you you understand it you could make a script or a computer program that just iterates and whether it experiences that in an anxious way kind of a second question but at least from that functional side it's it's uh as if ruminating that's the paper here yes thank you yeah so um let's see here they actually have a model of the connective inference terms so let's see where it starts and it's actually not so simple um okay so what they want to explain is basically how childhood male treatment leads to borderline personality disorder and this is important for my trans diagnostic marker of reduced emotion regulation because uh bad emotion regulation is the symptom of borderline personality disorder what they say is okay let's see that we have two relevant factors here a strong negative belief about the self and others it's also typically known for uh Borderline Personality Disorder so we have high Precision to use active inference terms here or Bayesian terms to my prior beliefs that other people are bad and I am also like a loser that cannot do anything right in his or her life and this needs of course if the Precision of the prior is really high new information doesn't matter and so we have a Persistence of uh effects of hostility because if the others are all bad well I can be hostile towards them or fear and of sadness um and in addition to that also contributing to the temple fluctuations because we know that borderline is characterized by sometimes uh um like I really like my therapist as a borderline patients and after a while I really hate him and not a lot in between and this may come about due to like a high uncertainty of beliefs about consequences of action so here it's about policies so here my standard deviation is really broad uh and due to that like belief updating regarding policies gets fast and this is why I show like uh instability while at the same time also showing persistence and I show Persistence of my world view but instability of my behavior and um this is how they try to explain borderline personality disorder in active inference terminology but this is only me giving it a try I only read the paper once but this might be one way how to characterize like symptoms of emotion regulation problems in a conceptual model using Bayesian terminology awesome and also in this paper and in yours it's on the road map towards formalization not like that's the only end point but it's going from the empirical evidence and the ways of thinking and knowing in the clinician's office through the integrative Frameworks to the pros and to the graphics that connect different nodes and edges this figure and the ones that you drew they're not exactly a base graph so it's not exactly ready to be entered into a computer program but yes it's also not far from a base graph you have variables that could basically be variables in a base graph and so it it moves along and brings collaborators and so on as uh in a kind of methodologically plural way there can be developments on multiple fronts and integrate the neural Imaging and the predispositions and then the treatment modalities in in a way that's not just like following the causal chain of the etiology but rather approaching it almost like in a reverse problem-solving way starting with the clinician's awareness and View and first seeking to integrate and improve the sense making of the clinician even without a formal model yeah I would totally agree to this not much I can say about it I think it's totally right what you said there um maybe one thing uh in clinical experience at least in Psychology I don't dare to speak of other fields um but they are backward translation is really important so this is what you described right so going back from the clinical phenomenon to like uh more precise view for example in um uh informal modeling and um typically problems got tackled the other way around and this oftentimes doesn't work as well as backwards translation because typically problems got uh thought through by like um basic scientists that are really good at doing this formal modeling but don't model necessarily the right parameters from our view at least because they lack the clinical experience and so both parties working together is really important like guys like us um bring in the important parameters or at least where we think they could be important from clinical practice and the more mathematically inclined can model them and so there could be a fruitful collaboration between so start with backward translation then do forward translation then do another loop of backward translation and I guess this is a fruitful way of advancing a Psychiatry and psychology wow a related question to what extent either today or in the future do you think patients would be clued in to some of this lexicon and ontology for example is it enough to just say hey here's the new recommendation here's the new exercise plan or is it explicitly oriented around communicating Within These ways of talking about belief uncertainty information and so on I think it can be done with patients and it should be done to a certain extent I think it's um not plausible to teach them all the mass and stuff like this is will not be possible for at least a lot of them of course sure if you have a mathematics professor patient but it would be easy for him but or her um but in most cases no not that but you could talk about Precision you could talk about surprisal you could talk about uh hierarchical learning and I guess it's important to do this you could simplify models into like uh vicious circles um one could easily draw a simple vicious circle with the help of our elimination model and uh teach them or the patients a few patients um adjacent words at least to better understand their condition and I think this is important because psycho education actually is proven to help right the more you know about the condition the higher your chances of getting better are in in a lot of cases and so I guess yes you could could teach them at least base rules and a few simple things not doing a lot of math but like drawing a few graphs and try to teach them the basics would be a good idea because I guess everybody is can get what precise information is and what uh like belief updating how it functions in principle this is teachable to most of us yeah one example that from the talk you mentioned like with the loss of an important person in our life our social or our emotional support policies are going to have to be updated if there was some Source we went to for information and that Source no longer became available we would have to change our epistemic foraging and of course this exists in the social space and so when life changes when our generative model or when our generative process our Niche changes we should expect the unexpected we and and so it's a totally um normal surprising but expected surprise event opens up the discussion to like there will be beliefs that will be changing there will be new policies or policies that you didn't previously take maybe now is the time to try them out and there's going to be volatility as you're learning the consequences of action yes so stick with it and try something three times before saying that it doesn't work um even without appeal to mathematics that's very powerful and I think the way that uh somebody might say well in this double-blind study these two groups this treatment that we're going to go with it's been shown that in this context these two groups this group did better and no one says well slow down you know the t-test used a lot of mathematics I'm not a statistics Professor so people accept the epistemology of the group differential statistics and of diagnostic statistics even though one could do 100 phds in theoretical statistics and in computational statistics and so just speaks I think to the early stage when the field is even perceived as being math heavy because we're in such a methodological developmental stage yes but there will be layers and ways of working that will not mention any more technical detail than someone just saying the double-blind study has provided evidence for x yeah I can say two things about it the one thing is a philosophical thing um Daniel dinette talked about free-floating rationales right so a free-floating rationale in case you don't know is like when Lions go for gazelles it's the example that he likes to give the gazelles to these showboating types of jumps and the Lions typically don't go after the gazelles that can show this Behavior but go after the injured ones that cannot do this anymore right but we can be quite certain that neither the lion nor the gazelle knows what they are doing but it works without knowledge and this is what he calls a free-floating rationale there is a rationale behind the behavior of the lion but he doesn't need to know it in order to take advantage of it and this is the case for Science and this is also the case for patients you don't need to know every detail of a mathematical formalism in order to grasp some of the concepts to advance you um your understanding of the world using these Concepts but it of course it helps but uh for patients it's the same and for us it's the same and I think a free-floating rationales are really important concept to describe why a lot of the times you don't need to know the exact formalisms and another idea that I want to give is more like a science of making Niche constructions more applicable to the mainstream and this is well at the beginning it's hard because you need to program every little bit yourself there are no toolboxes nothing to help you and not really books just a few experts but after a while books come out after a while you get toolboxes that actually simplify a lot of the process you don't you don't need to write the helper functions yourself you can use the toolbox in a lot of cases and so after a while it becomes more widely applicable in science and maybe at some day I don't know SPSS or something one of the statistics softwares maybe not SPSS has like a way to click through active inference algorithms let's see but then in principle everybody can use it when you have new problems because people will use it without actually understanding it but um I would predict that it gets simpler and simpler to to use and I would say overall this is a good thing because this allows a lot more scientists to jump in on the Endeavor thanks yeah great points one of your quotes the short one was existence precedes essence and it reminded me of a 2015 paper of friston and Mathis which is entitled I am therefore I think and that of course is a little bit of an inversion on I think therefore I am which puts rumination before the horse or the cart before the horse or I'm not exactly sure but I I think therefore I am it actually puts thinking and ruminating first whereas this existentialist quote that you provided Concords well with certain ways of thinking about reflexive systems in active inference so what did you mean with the introduction of that quote and how does that existentialist angle play out in the clinical setting questions disguised this one I stuck with the existentialist uh worldview and clinical settings um well I think it's important for clinical science because and more so for clinical practice because we need to believe in a certain type of Liberty in order to help patients if we believe everything is predetermined by I don't know a brain chemistry even if it's true this will make changing your behavior hard because you can always say well I cannot change my behavior it's all in my I don't know genes or what type of biological reductionism you like so I guess we need these existentialist idea that yours you yourself are in control of your life in order to make therapy work right so we need to at least have a compatible uh hard English words like this even if determinism is true we need to have uh it's combined with some sort of of Liberty or belief in Liberty and I guess this is where active inference jumps in because active inference this allows a formal description about self-control and then we have a definition of of uh of acting in the world and learning new things that doesn't require like fairy tales uh and it's compatible with determinism um and uh with this word for you you have a worldview that is um on one hand is compatible with determinism and on the other hand allow us for a concept like self-control agency policy learning all the things and so it can also be used for psychotherapy interesting I hope this answers the question but this is what I can say yeah there's so much uh depth obviously many threads and traditions in in the clinical setting and that full stack from the Manifest behavioral on through the cognitive and psychological but verging or anchored in or descended from the metaphysical thoughts that's a full spectrum area that seems like it must be navigated in the clinician's education but even in like the one hour that you're spending with somebody I imagine that it could probably uh oscillate a lot in where you're focusing from here's how to tie your shoe to here's why things matter abstractly they asked more philosophy especially practical philosophy so therapy then one would think because patients bring it up all the time right I mean one of the problems of depression if if it's going to be really severe is this for example nihilism kicks in and um so nothing makes sense anymore and then as you said why should anything matter at all then becomes an interesting question and the pressing question for them what's your go-to response why does anything matter oh it's a hard question I would try to get um do you always thought that nothing matters what was different when something mattered for you how was your life then um and what changed why you came to these conclusions now and how do you feel when you think about your old worldview versus new worldview so I would try to challenge their world for you by their own worldview a few years ago this is not always working because you have patients that have actually gone through a lot of difficult times in their lives but typically like their worldview was not always depressed yes that seems to do multiple things by responding to a question with a question it puts an active posture back into the conversation so at the very least policies can be selected that aren't rumination and consequences could be learned even if there's going to be a a higher low Precision on that learning at least it like opens the door to movements whereas just well person X said this or that that response may land or Inspire or not but it reifies the pacification and therefore may not uh um kind of like a inactive layer yeah promote the movement um yeah yeah I just wanted to say yes awesome everything's the same um and if you try something like a person XY that says this is the reason why your life works this will in most cases not work at all they will say something like but this person doesn't go through that many hardships that I did what is oftentimes true um or something like that right so often it's better if the ideas are sampled from their own generative models from a lost part of the distribution so to say and not from outside typically you have to to try to get them to think out of their own world for you and not of your work you or someone else as well if you were this seldom functions um so another interesting area you mentioned the problem difficulty complexity time uncertainty estimations so kind of like a cognitive task analysis in the CBT setting where different tasks are estimated again along these axes of how challenging is that for you or how complex is that task seem to you um how does that really like play out because it really seems like something in the email inbox or in writing a paper with some improved signposting or learning even around which tasks are hard which tasks are clear which tasks are um threatening our well-being those kinds of metadata on policy selection and being able to see that could help us select better policy and not to change our preferences which is how sometimes behavioral change is approached by changing what people are supposed to want but by bringing this kind of cybernetic clarity around action selection to support people in different contexts making the decisions that can then credibly be said to be more in alignment with what they do want yeah I mean what is typically already been done is um to let patients guess how uh how they would rate the probability of succeeding right so when we do this behavioral therapy they are typically asked okay um what do you think how probable is it that you're gonna go there until next week um and it's a rule of thumb by clinicians it has no base in science but we say if it's 80 or higher probability one should aim for it if it's lower it's too difficult um but if it's trivial if it's 99.9 probability uh it's also not worth um typically pursuing at least not always um but I guess a problem with that is that I don't know a solution for this is my subjective probability right how I think how probable it is I don't have a good grasp in all cases on how difficult it really is for me because there are all sorts of biases that would keep me from making such estimations I don't have a solution for that but what would be good is to have more variables that we could uh estimate in order to actually have more information on how difficult the task is but for most cases we haven't and if when we have them we know that What patients think how difficult it is and how difficult it really is hasn't the best correlation sure it's positively correlated but how good I think my short-term memory for example is and how good it really is it's not the same and this is a hard to solve problem for clinical science I have to think about it longer but it would giving patients tulo to actually improve their probability estimates would be great but I guess it's also hard to do due to all these things that you said before right if the problem is complex multi-layered and has a lot of areas and good luck actually finding um a way to estimating probabilities but we could start for simple problems and maybe this helps patients but I have to think about it more to be honest it's it's not easy to find better Solutions there hmm yeah even just on paper or digitally journaling but then eventually potentially integrating with the kinds of formal models we've been discussing it can be really nice to know hey there's a lot of stuff that you said had a below 50 chance of happening and nine out of ten times you've done that so maybe next time you're in the zone of uncertainty just go for it yeah and or just this Pat you tend to systematically overestimate your confidence or um you um you over underestimate how much other people will appreciate this kind of consequence and maybe writing three thank you notes a day would be a great use of 20 minutes um yeah these are all really good ideas um and it's this is also a little is like tracking certain behaviors like uh an easy example would be aggressive behavior for example counting how much like verbal aggression was shown or physical aggression was shown over the days and then like uh look what the expectation of the patient was and comparing those two and there you could do like basic belief updating in in a simplified form yeah yeah this would be a great idea always then possible I would say when the behavior is clear enough and easy enough to count right as soon as if it's getting more unclear uh this will not work but it could work for a lot of problem behaviors like self-harm aggression lying in bed like Forex in depression something like that so it makes me think about that contrast that you provided between reward oriented Frameworks and which would struggle to describe rumination as it's not in the short or medium or long term necessarily oriented with rewards so the only explanation becomes well something is off in the reward function which is kind of tautological and so that's a little bit of a of a local and a constrained explanation whereas when we see policy selection external overt movement in the world as well as internal covert policy selection like attention and and mental action when we see those actions as being oriented around reduction of free energy which is bounding surprise and therefore based upon what is likely not what is necessarily the most rewarding um it it opens up Dot um View to again say how many of this Behavior do you expect to happen in the next week yes in a dream world we would aim for zero of events of this number you know we'll aim for you to be full-time employed by next week that sounds awesome but will we honestly expect in one two three four five weeks and then we're going to actually be looking back in and then it's um instead of maybe feeling like um well you're gonna leave the port and then you're gonna have to sail across an ocean it can be more like there are checkpoints we're looking back at the buoys that we passed we have the next three buoys that we're looking towards we can do smaller Corrections and there will be volatility so there's going to continue to be surprised we're not going to exactly hit every Milestone per se but if we are on this trajectory we're going to be moving essentially as much as we possibly could in the right direction and spending less time just um hypothesizing about like a total counterfactual world where the issue is already resolved yes and this is a method for um metaphor that is also used in therapy quite often right the one of the mountain climb bar and the mountain climber typically needs a guide if he's a newbie and what Psychotherapy can do is like provide like a few a good way over the mountain so to say provide a good place for um for resting doing the next step doing the next step doing the next step and yeah this is really resonating with what you said namely we have to take a big problem and cut it into little problems that we can solve one after another so um it becomes plausible to actually solve it yes I would totally agree yeah who an emotional flanker tasks not exactly sure what those would be but it just made me think of our policy selection throughout the day and all of the fires we have to put out or all of the bubbling soups that we have to tend to with thirst and rest and breathing and physical exercise and all of these different aspects to our our cognitive and emotional well-being it's like there's many things to attend to which is yes why our attention is so a variable as a feature it's not a bug so the fact that we have exactly pathological or not just neurodiverse states that involve a fixation of attention and also the inability to fixate attention it just starts with our experience of the day and of our attention to our body and our surroundings and to the bigger questions that we know impinge on us but also we can't necessarily um have the geopolitics and the climate change solved just by thinking about it but we also don't want to neglect what are important issues and then seeing strategies and having the discussion about strategies and tactics in that attentional space instead of in the reward space not to um you know bash or anything like that but that conversation is about well what should we value more or less and then why are these values uh translating effectively or not into Behavior whereas the attention and surprise angle is like being more in the game and your local view instead of like watching the game and just saying well I want this to be the outcome exactly yeah I want to say a little bit about this attention focus and attention spreading that you mentioned um I almost said it in the talk but I say it now like for example ADHD is not necessarily like uh disorder in the sense that it's a dysfunction right so it can also be described as maybe better described as a neurodivergent um property of some human beings um so they shift attention more easily and this is depending on the environment a good thing because if you stuck your attention at one place too much well you might not notice what is going around in your surroundings so you have to be able to focus away from certain points of attention and you want to have in a group individuals that have really sticky attentions so they can focus on a task really good but you also want to have the other ones that have like really floating attention spans in order to the new opportunities or see Danger and so I think this can be really easily explained by evolutionary theory right so we have uh uh variance over the population and this variance is not a bug it's a feature and on the more extreme ends of the variance you find uh phenotypes that are problematic from time to time but depending on the surroundings always depending on the surroundings right so on the one hand you might have something like ocd-like behavior and on the other hand something like more ADHD like behavior and it can even jump between the two interestingly but this is not necessarily always bad it really depends on your environment and that humans differ with respect to this parameters is also not a back but a feature of evolution I would say and also a feature of our learning history of course totally agreed for individual and Collective behavioral strategies they're often discussed as if they were context Independence when there are ant colony foraging strategies that work in the desert and they work not as well in the jungle so just like it is with our attention there's attentional surroundings that support this kind of work but not that kind of work or well-being so being able to talk about the generative model and the generative process and their engagement in a rich way I guess that's what brings us together but it's also what feels like a powerful place to re-gather and remember a lot of the importance qualitative threads and empirical threads in this important clinical area and also move forward in a way that's not just adding like another measurement onto the pile because a lot of these models as you continue to develop their formality and ability to integrate data they will provide a new view on previous findings and patterns that may have been taken in the t-test reward learning world to be conclusive for this or that will be shown to be just two darts thrown into a huge neurodiversity space and there might be other studies or sets of studies where what was treated as being all over the place are now like perfectly in line until a really coherent story about for example how strong negative beliefs about the self and others or high uncertainty of beliefs about the consequences of action end up helping us just do what we do yes yeah totally agree it's maybe it's interesting there's even reconceptual elevation going on like uh thinking about Psychopathology and more like evolutionary terms like uh Stefan Hoffman for example and Stephen Hayes Stephen Hayes is known as the founding father of um acts so acceptance and commitment based therapy and those two guys have a new therapy idea where we describe um Psychopathology on a pragmaticist basis so we Define psychopathology as behavior that doesn't fit anymore the current circumstances so there you have the context in right and I think it's clever to do it more that way because for example PTSD like Behavior well if you're a soldier in a in a really bad region of the world this is not Psychopathology this is allowing you to survive the problem is of pthd that the war comes home with you or traumatic event comes home with you and now your behavior doesn't fit the environment anymore right and I think this is a neat way of describing a lot of Psychopathology as maladaptive behavior in The evolutionary sense and typically it was adaptive in the past um like also what this like borderline personality disorder slide that I have on the screen right now it's the same because it was adaptive in the past to behave like this in in some circumstances in a bare childhood so to say but it's not anymore but it's continued despite being ineffective and I guess this is a good definition for a lot of Psychopathology not for all I I would say for example there is really Psychopathology where you have broken inference or to say and not just if for example in in psychosis in some cases I would say yes uh you have more like brain pathology you have more really broken inference in the rules of active inference don't work anymore so to say but I don't think that the rules of active inference don't work in rumination they work perfectly fine but um the organism has placed himself in uh in a situation where they don't learn anything new by applying the same rules that everybody else applies so it's broken models not broken inference and I guess this is true for a lot of psychopathology um so what are your next moves with uh this line of research you mentioned some of the papers what would you like those papers to show or who would you like them to speak to and show what um maybe I stuck with the literature uh review that we are planning I would like to get to know what uh the the literature so to say says about active inference informed treatments this is what I would like to find out I don't have a lot of hypotheses I want to screen the literature and find out whether we can synthesize them into an active inference informed model of what is psychotherapy doing and I would like to speak this model to a clinical audience in the first place again so I want uh the clinical population to learn more about active inference I know that psychiatrists sometimes know about active inference literature but psychologists at least not clinical psychologists typically don't know a lot about it and I would like to change this a little bit at least and afterwards um the next logical step would be to implement some of the inside intellectual therapy programs but this is something for more of the future but it would be interesting whether you can really have something like an active inference informed therapy but now we're talking more like five to ten years do you have any final thoughts or questions and then I will close with a short quote to fire back yes sir no I don't have any final remarks or questions I would like to thank you for giving me the opportunity to speak and now I want to resolve my uncertainty regarding the quote great the quote is from Henry David Thoreau and it's drawing a connection between the paths that we ruminate on that we walk upon and travel upon and then the paths in the mind so I thought it was Africa so throw row I had not lived there a week before my feet wore a path from my door to the pond side and though it is five or six years since I tried it it is still quite distinct it is true I fear that others may have fallen into it and so helped to keep it open the surface of the Earth is soft and impressible by the feet of men and so with the paths which the Mind travels how worn and Dusty then must be the highways of the world how deep the ruts of tradition and conformity awesome thank you for having me till next time thanks a lot see you bye